

Def. A map  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{C} : (m, n) \mapsto a_{mn}$

is called a Double Sequence.

observe that: define a set  $X \stackrel{\text{def}}{=} \{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{N}\}$ .

For a Continuous map  $f: X \rightarrow \mathbb{R}$ , a sequence  $a_n = f(\frac{1}{n})$  converges to  $f(0)$ . Conversely, for a Convergent sequence  $\{a_n\}$ , the map  $f$  defined on  $X$  as

$$f(0) = \lim_{n \rightarrow \infty} a_n, \quad f(\frac{1}{n}) = a_n \quad (n \in \mathbb{N})$$

is Continuous.

Thm. Let  $\{a_{mn}\}$  be a double sequence. If  $\{a_{mn}\}$  satisfies the followings:

1) For each  $m \in \mathbb{N}$ ,  $a_{mn} \rightarrow b_m$  as  $n \rightarrow \infty$

2) For each  $n \in \mathbb{N}$ ,  $a_{mn} \rightarrow c_n$  as  $m \rightarrow \infty$ .

3) For any  $\epsilon > 0$ , there exists  $N \in \mathbb{N}$  s.t.  $m \geq N \Rightarrow \sup_{n \in \mathbb{N}} |a_{mn} - c_n| \leq \epsilon$ .

Then, 
$$\lim_{m \rightarrow \infty} b_m = \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} a_{mn} = \lim_{n \rightarrow \infty} \lim_{m \rightarrow \infty} a_{mn} = \lim_{n \rightarrow \infty} c_n$$

Proof. For each  $m \in \mathbb{N}$ , define  $f_m$  on  $\{\frac{1}{n} \mid n \in \mathbb{N}\}$  as  $f_m(\frac{1}{n}) = a_{mn} \quad (n \in \mathbb{N})$

Then, we can observe that

1)  $\Leftrightarrow$  For each  $m \in \mathbb{N}$ ,  $\lim_{t \rightarrow 0} f_m(t) = b_m$

2)  $\Leftrightarrow f_m$  converges pointwise to  $f$ , where  $f(\frac{1}{n}) = c_n \quad (n \in \mathbb{N})$

3)  $\Leftrightarrow f_m$  converges uniformly to  $f$ .

Therefore, 
$$\lim_{m \rightarrow \infty} b_m = \lim_{m \rightarrow \infty} \lim_{t \rightarrow 0} f_m(t) = \lim_{t \rightarrow 0} \lim_{m \rightarrow \infty} f_m(t) = \lim_{t \rightarrow 0} f(t) = \lim_{n \rightarrow \infty} f(\frac{1}{n}) = \lim_{n \rightarrow \infty} c_n.$$

Lemma. Let  $f_n: X \rightarrow Y$  be continuous functions s.t.  $f_n \rightarrow f$ .

If a sequence  $\{x_n\}$  in  $X$  converges to  $x \in X$ , then  $\{f_n(x_n)\}$  converges to  $f(x)$ .

Proof. Since  $f_n \rightarrow f$  and  $f_n$  is continuous,  $f$  is continuous.

Given  $\epsilon > 0$ , there exists  $N \in \mathbb{N}$  s.t.  $n \geq N_1 \Rightarrow \sup_x |f_n - f| < \frac{\epsilon}{2}$ .

Meanwhile, for these  $n \geq N_1$ , there is  $N_2 \in \mathbb{N}$  s.t.  $m \geq N_2 \Rightarrow |f_n(x_m) - f_n(x)| < \frac{\epsilon}{2}$ .

Now, for  $k \geq \max\{N_1, N_2\}$ ,

$$|f_k(x_k) - f(x)| \leq |f_k(x_k) - f_k(x)| + |f_k(x) - f(x)| < \epsilon. \quad \square$$

Thm. Let  $\{a_{mn}\}$  be a double sequence s.t.

1) For each  $m=1, 2, \dots$ ,  $\sum_{n=1}^{\infty} |a_{mn}| = b_m$

2)  $\sum_{m=1}^{\infty} b_m$  converges.

Then,  $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{mn} = \lim_{N \rightarrow \infty} \sum_{m,n=1}^N a_{mn}. \quad \dots (*)$

Proof. Let  $X = \{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{N}\}$ , and for each  $m \in \mathbb{N}$ , define

$$f_m: X \rightarrow \mathbb{R}; x \mapsto \begin{cases} \sum_{j=1}^m a_{mj} & \text{if } x=0 \\ \sum_{j=1}^n a_{mj} & \text{if } x=\frac{1}{n}, n \in \mathbb{N}. \end{cases} \quad \left( \text{That is, } f_m\left(\frac{1}{n}\right) = \sum_{j=1}^n a_{mj} \right)$$

Weierstrass M-test  $\downarrow$

and let  $g_n = \sum_{m=1}^n f_m$ . Since  $|f_m| \leq \sum_{j=1}^{\infty} |a_{mj}| = b_m$  and  $\sum b_m$  converges,  $g_n$  converges uniformly.

And, each  $f_m$  is continuous, thus  $g_n = \sum_{m=1}^n f_m$  is continuous, thus  $g = \lim_{n \rightarrow \infty} g_n$  is continuous, therefore

$$\lim_{n \rightarrow \infty} g\left(\frac{1}{n}\right) = g(0) = \lim_{n \rightarrow \infty} g_n\left(\frac{1}{n}\right).$$

g-Conti.  $\uparrow$   $\lim_{n \rightarrow \infty}$   $\uparrow$  Lemma.

Moreover,

$$\lim_{n \rightarrow \infty} g\left(\frac{1}{n}\right) = \lim_{n \rightarrow \infty} \sum_{m=1}^{\infty} f_m\left(\frac{1}{n}\right) = \lim_{n \rightarrow \infty} \sum_{m=1}^{\infty} \sum_{j=1}^n a_{mj} = \lim_{n \rightarrow \infty} \sum_{m=1}^{\infty} \sum_{j=1}^{\infty} a_{mj} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{mn}$$

$$g(0) = \sum_{m=1}^{\infty} f_m(0) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn} \quad \text{In finite sum, convergences preserved limit.}$$

$$\lim_{N \rightarrow \infty} g_N\left(\frac{1}{N}\right) = \lim_{N \rightarrow \infty} \sum_{m=1}^N f_m\left(\frac{1}{N}\right) = \lim_{N \rightarrow \infty} \sum_{m=1}^N \sum_{j=1}^N a_{mj} = \lim_{N \rightarrow \infty} \sum_{m,n=1}^N a_{mn}.$$

$\square$

Note: If  $(*)$  holds, then we say that sum of  $\{a_{mn}\}$  can be exchanged.

Thm. If  $\{a_{mn}\}$  is positive, then sum of  $\{a_{mn}\}$  can be exchanged.

Proof. For any  $M, N \in \mathbb{N}$  s.t.  $M \leq N$ , since  $a_{mn} \geq 0$ ,

$$\sum_{m=1}^M \sum_{n=1}^N a_{mn} \leq \sum_{m,n=1}^N a_{mn}$$

Taking limit to the both side, then

$$\sum_{m=1}^M \sum_{n=1}^{\infty} a_{mn} = \lim_{N \rightarrow \infty} \sum_{m=1}^M \sum_{n=1}^N a_{mn} \leq \lim_{N \rightarrow \infty} \sum_{m,n=1}^N a_{mn}$$

Since  $M$  was arbitrary,

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn} \leq \sum_{m,n=1}^{\infty} a_{mn}.$$

And, the converse inequality is given by: for any  $N \in \mathbb{N}$

$$\sum_{m,n=1}^N a_{mn} \leq \sum_{n=1}^N \sum_{m=1}^{\infty} a_{mn}.$$

Corollary. If  $a_k \geq 0$  for all  $k \in \mathbb{Z}$ , then

$$\lim_{n \rightarrow \infty} \sum_{k=-n}^n a_k = \lim_{n \rightarrow \infty} \lim_{m \rightarrow \infty} \sum_{k=-m}^n a_k = \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \sum_{k=-m}^n a_k.$$